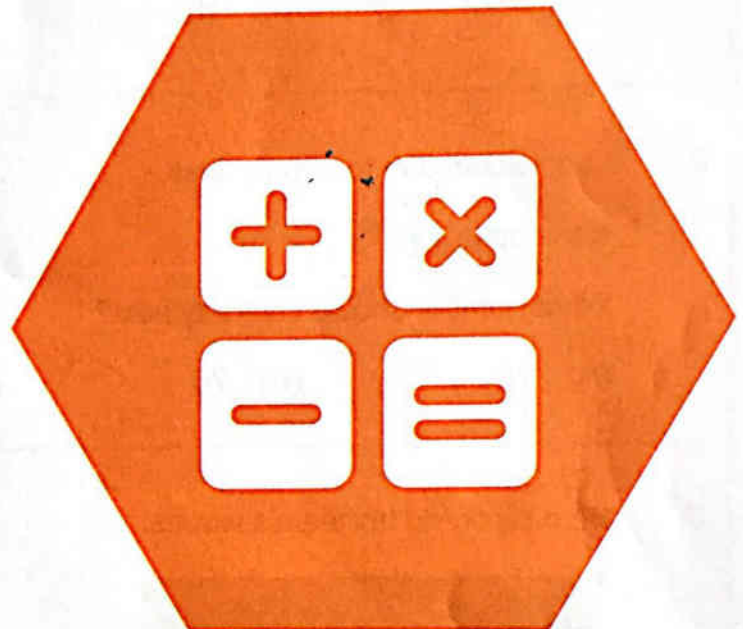




UNSW Global
AUSTRALIA

INTRODUCTORY PAPER



2016 ICAS

International Competitions
and Assessments for Schools

MATHEMATICS

Educational Assessment Australia
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DO NOT OPEN THIS BOOKLET UNTIL INSTRUCTED.

30 QUESTIONS

TIME ALLOWED: 35 MINUTES

STUDENT'S NAME:

Read the instructions on the **ANSWER SHEET** and fill in your **NAME, SCHOOL** and **OTHER INFORMATION**.

Use a 2B or B pencil.

Do **NOT** use a pen.

Rub out any mistakes completely.

You **MUST** record your answers on the **ANSWER SHEET**.

Mark only **ONE** answer for each question.

Your score will be the number of correct answers.

Marks are **NOT** deducted for incorrect answers.

There are **30 MULTIPLE-CHOICE QUESTIONS** (1–30).

Use the information provided to choose the **BEST** answer from the four possible options.

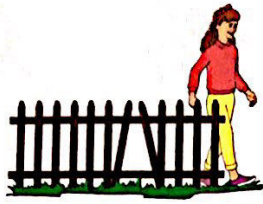
On your **ANSWER SHEET** fill in the oval that matches your answer.

You may use a ruler and spare paper.

You are **NOT** allowed to use a calculator.

Scanned by CamScanner

1. Which of these shows Lucy climbing over the fence?



(A)



(B)



(C)



(D)

2. Fadi is counting to 100 by fives.

He is up to 75.

What number should Fadi say next?

- (A) 70 (B) 76 (C) 80 (D) 85

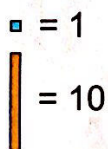
3. Pete recorded his team's results.

Won	Lost

How many games did Pete's team win?

- (A) 2 (B) 3 (C) 4 (D) 5

4. Lisa is using blocks like these to make numbers.



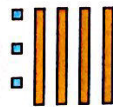
Which of these makes the number 35?



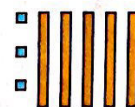
(A)



(B)



(C)

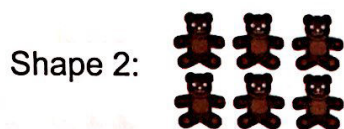


(D)

5. Which list gives three months, from the same year, in the correct order?

- (A) January, August, April
- (B) January, April, August
- (C) April, January, August
- (D) April, August, January

6. Ying arranged some bears in a pattern.



Which of these was Shape 4?



(A)



(B)



(C)



(D)

7. Which of these numbers is between 65 and 72?

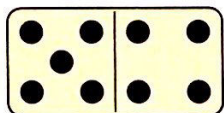
- (A) 59
- (B) 64
- (C) 67
- (D) 73

8. Jim is matching dominoes by adding their spots together.

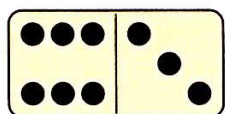
For example:



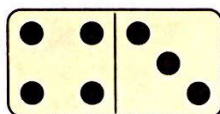
Jim picked up this domino.



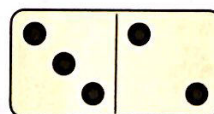
Which domino matches it?



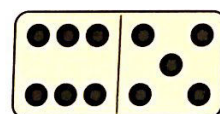
(A)



(B)

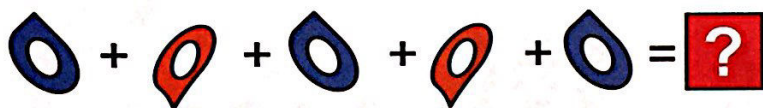


(C)



(D)

9.



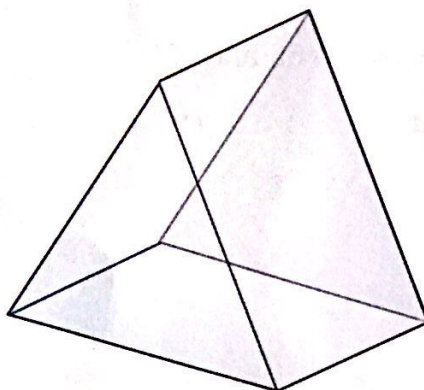
(A) 12

(B) 14

(C) 15

(D) 29

10. How many faces does this object have?



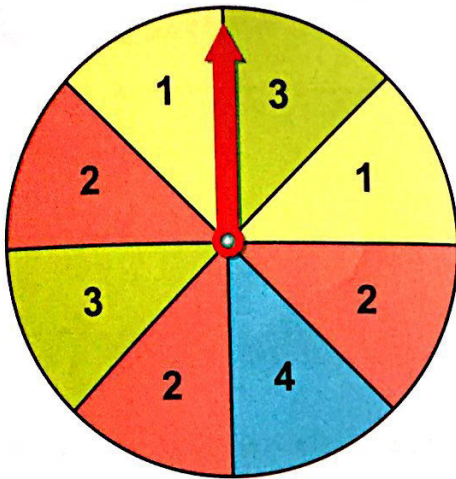
(A) 3

(B) 4

(C) 5

(D) 6

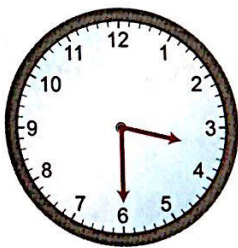
11. Lin has this spinner.



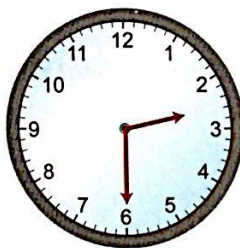
On which number is the spinner most likely to land?

- (A) 1 (B) 2 (C) 3 (D) 4

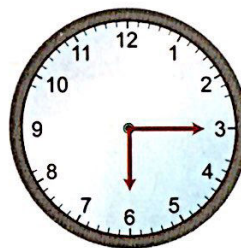
12. Which clock shows half past three?



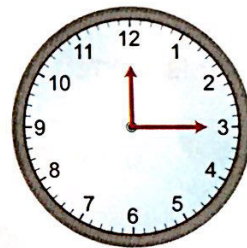
(A)



(B)



(C)



(D)

13. Jim drew some pictures.

Which of these is **impossible** for him to draw?

- (A) a cat with three legs
(B) a square with five sides
(C) a table with five legs
(D) a car with three wheels

14. Mark wrote this number sentence.

? + 3 = 4 + 7

What number must **?** be?

- (A) 14 (B) 11 (C) 9 (D) 8

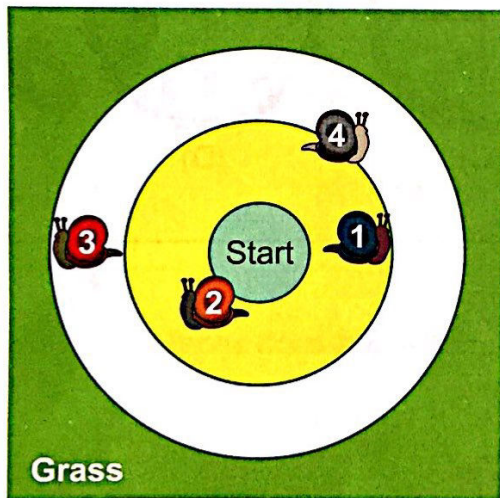
15. Cai had 30 jelly beans.



He shared the jelly beans equally among his 5 friends, keeping none for himself.

How many jelly beans did each friend receive?

- (A) 5 (B) 6 (C) 25 (D) 35
16. Four snails started in the centre circle. The picture shows their positions after 5 minutes.

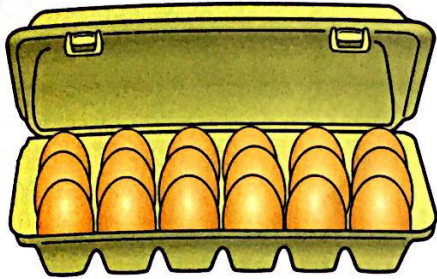


Which list gives the order of the snails, from the closest to the grass to the furthest from the grass?

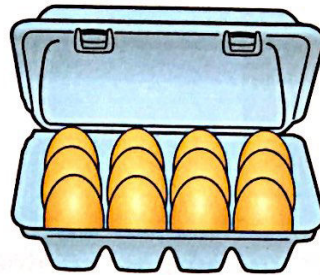
- (A) 3, 4, 1, 2 (B) 3, 1, 4, 2 (C) 1, 3, 4, 2 (D) 1, 4, 3, 2

17. Jude has cartons in two sizes.

A large carton holds 18 eggs.



A small carton holds 12 eggs.

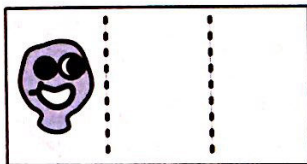


Jude filled 2 large cartons with eggs.

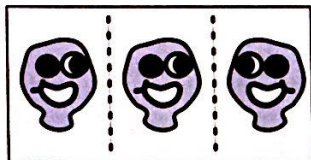
How many small cartons could Jude fill with these eggs?

- (A) 6 (B) 5 (C) 4 (D) 3

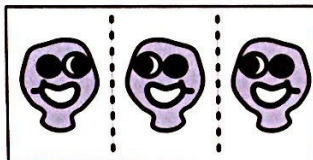
18. Sarah used this face to make a pattern. She flipped it over the dotted lines.



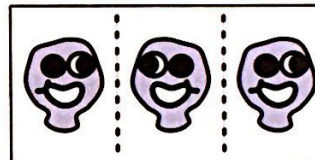
Which picture shows the result?



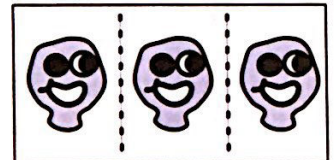
(A)



(B)

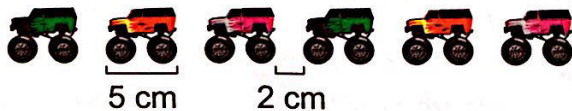


(C)



(D)

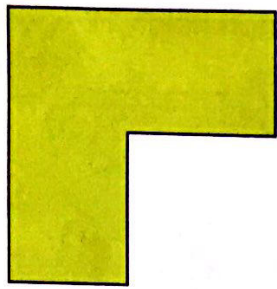
19. Tina lined up 6 toy cars. Each car was 5 centimetres long. She left a 2 centimetre gap between each car and the next.



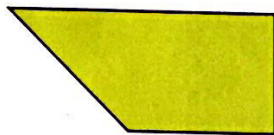
How long was this line of cars?

- (A) 30 centimetres (B) 32 centimetres
(C) 40 centimetres (D) 42 centimetres

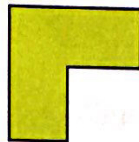
20. Ajay cut this shape in half.



What was the size and shape of each piece?



(A)



(B)

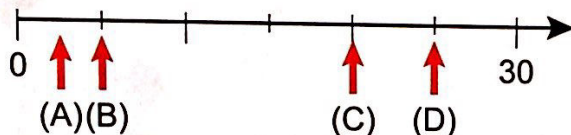


(C)



(D)

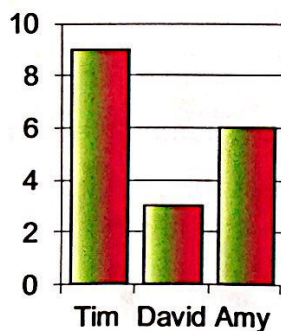
21. Which arrow points to 5 on this number line?



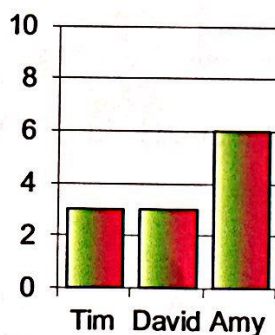
22. Tim, David and Amy picked some apples.

David picked 3 apples. Amy picked twice as many apples as David picked. Tim picked 3 more apples than Amy.

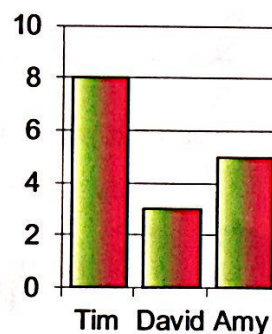
Which of these graphs shows the number of apples they each picked?



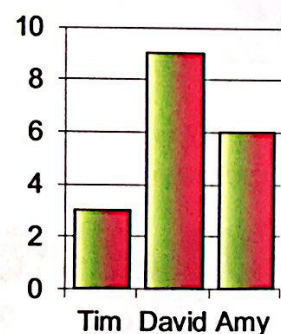
(A)



(B)

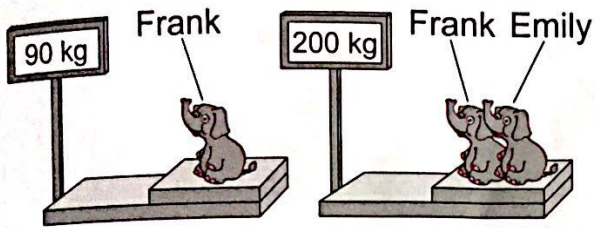


(C)



(D)

23. Emily and Frank are two baby elephants.



What is Emily's mass?

- (A) 90 kilograms (B) 110 kilograms (C) 190 kilograms (D) 200 kilograms

24. Pete made up this number pattern.

37	34	31	28	25	...
----	----	----	----	----	-----

What are the next three numbers in the pattern?

- (A)

21	18	15
----	----	----

 (B)

22	17	14
----	----	----

 (C)

22	19	16
----	----	----

 (D)

24	23	22
----	----	----

25. The graph shows the number of goals scored by the school basketball team over four weeks.

Week	Number of goals
1	     
2	    
3	  
4	   

 = 3 points

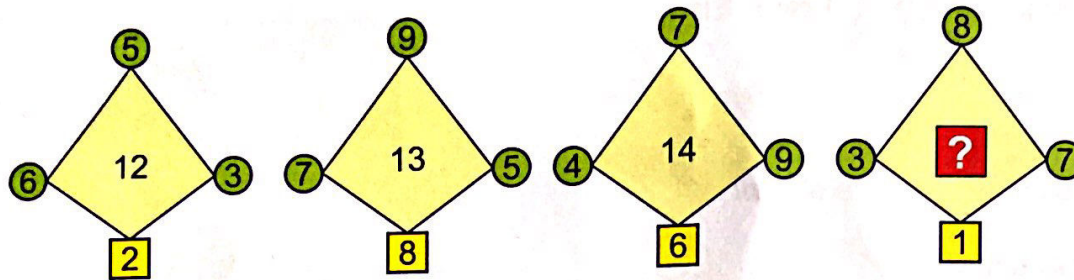
 = 2 points

 = 1 point

In which week did the team score the most points?

- (A) 1 (B) 2 (C) 3 (D) 4

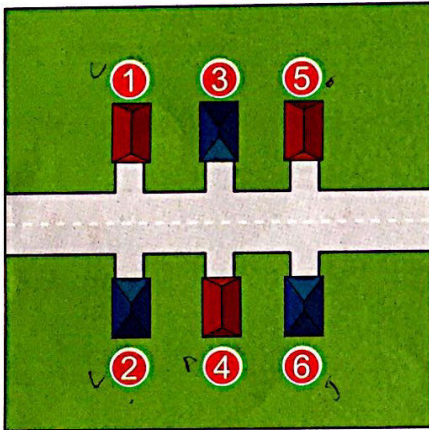
26. In each diagram, the outside numbers are used to calculate the middle number.



What number must ? be?

- (A) 13 (B) 15 (C) 17 (D) 19

27. Colin, Aditi, Jake, Ben, Samira and Lin live in houses on the same street.



Samira lives in house number 5. Colin lives on the same side of the street as Samira.

Aditi lives in the house between Lin and Jake. Lin lives opposite Colin.

What is the number of the house that Ben lives in?

- (A) 4 (B) 3 (C) 2 (D) 1

28. Jane gets one free horse-riding lesson for every 5 lessons her mum pays for at Wild Riders Pony Club.

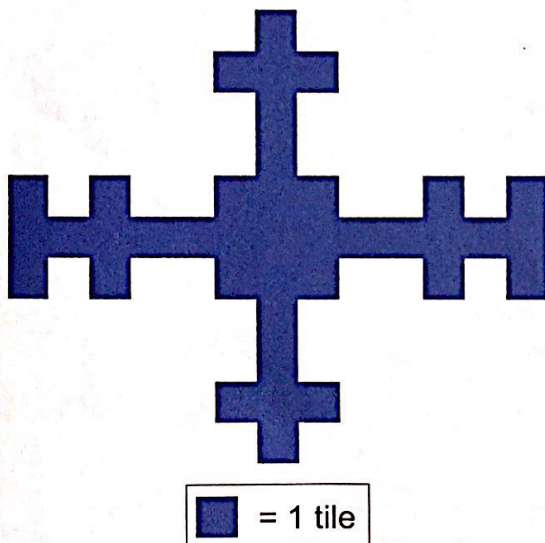
WILD RIDERS  Reward card					
1	2	3	4	5	Free
1	2	3	4	5	Free
1	2	3	4	5	Free

Jane's mum only paid for 32 lessons using these reward cards.

How many horse-riding lessons did Jane have?

- (A) 32 (B) 37 (C) 38 (D) 39

29. Aditi joined square tiles of the same size to make this shape. The tiles do not overlap.



How many tiles did she use?

- (A) 32 (B) 33 (C) 39 (D) 41

30. Debbie, Sue and Alice ran a race.

When Debbie had run 70 metres, Sue had run 40 metres and Alice was halfway between Debbie and Sue.

How many metres had Alice run?

- (A) 20 (B) 35 (C) 50 (D) 55

Acknowledgment

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PRACTICE QUESTION

What is the missing number?

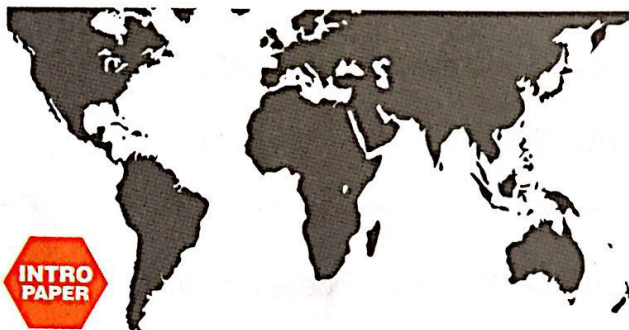
$$2 + 3 = \boxed{?}$$

- (A) 2
- (B) 3
- (C) 4
- (D) 5

Turn your answer sheet over and fill in the correct oval.

The following year levels should sit THIS Paper:

Australia	Year 2
Brunei	Primary 2
Indonesia	Year 3
Malaysia	Standard 2
New Zealand	Year 3
Pacific Region	Year 3
Singapore	Primary 1
South Africa	Grade 2



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2016

ICAS

Mathematics

1	D	16	B
2	C	17	D
3	D	18	C
4	A	19	C
5	B	20	A
6	D	21	B
7	C	22	A
8	A	23	B
9	D	24	C
10	C	25	D
11	B	26	C
12	A	27	B
13	B	28	C
14	D	29	C
15	B	30	D